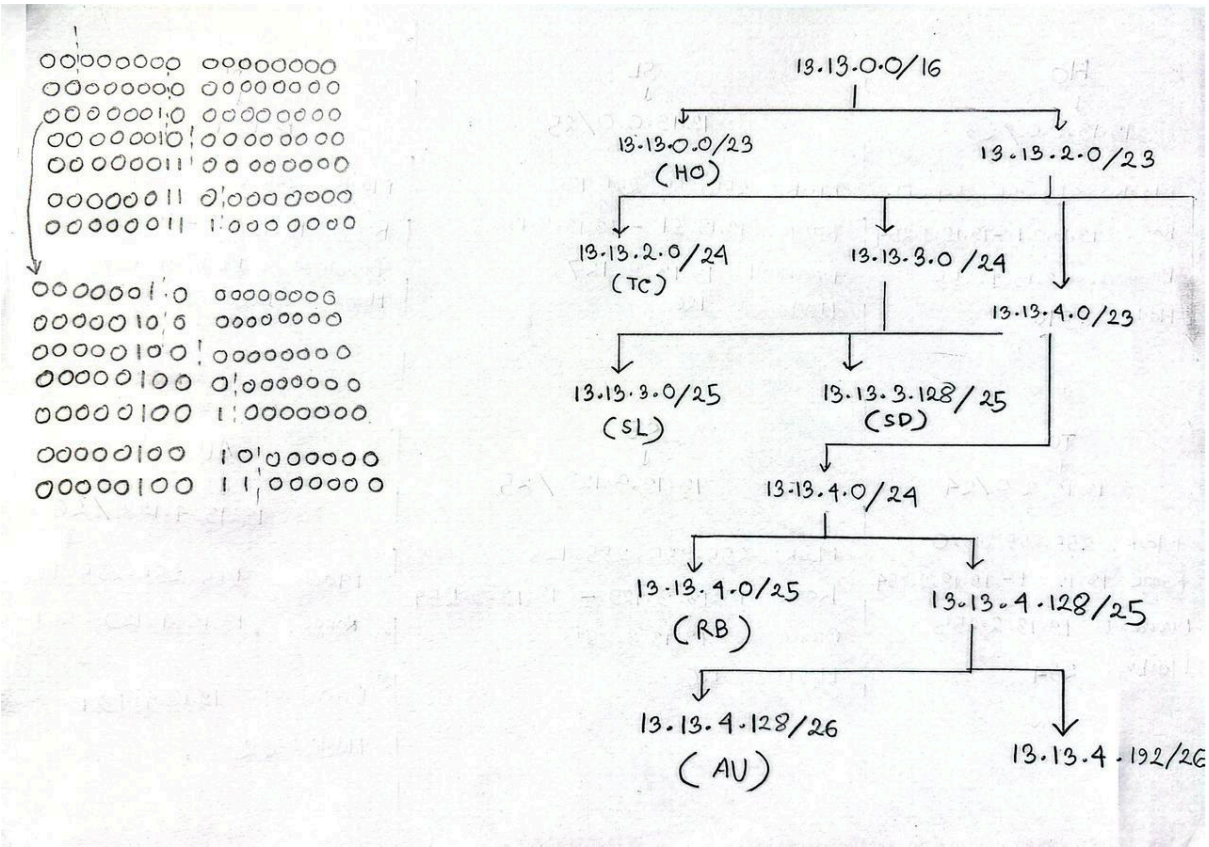
VLSM Tree



HO
↓
13.13.0.0/23

Mask: 255.255.254.0
Range: 13.13.0.1 - 13.13.1.254
Broadcast: 13.13.1.255
Hosts: 510

SL
↓
13.13.3.0/25

Mask: 255.255.255.128
Range: 13.13.3.1 - 13.13.3.126
Broadcast: 13.13.3.127
Hosts: 126

RB
↓
13.13.4.0/25

Mask: 255.255.255.128
Range: 13.13.4.1 - 13.13.4.126
Broadcast: 13.13.4.127
Hosts: 126

TC
↓
13.13.2.0/24

Mask: 255.255.255.0
Range: 13.13.2.1 - 13.13.2.254
Broadcast: 13.13.2.255
Hosts: 254

SD
↓
13.13.3.128/25

Mask: 255.255.255.128
Range: 13.13.3.129 - 13.13.3.254
Broadcast: 13.13.3.255
Hosts: 126

AU
↓
13.13.4.128/26

Mask: 255.255.255.192
Range: 13.13.4.129 - 13.13.4.190
Broadcast: 13.13.4.191
Hosts: 62

IP Addressing Table

Router Interfaces:

Device	Interface	IP Address	Mask	Purpose
HO-Router	Gi0/0	13.13.0.1	255.255.254.0	LAN Gateway
HO-Router	S0/0/0	13.13.5.1	255.255.255.252	Link to TC
HO-Router	S0/0/1	13.13.5.5	255.255.255.252	Link to SL
HO-Router	S0/1/0	13.13.5.9	255.255.255.252	Link to RB
TC-Router	Gi0/0	13.13.2.1	255.255.255.0	LAN Gateway
TC-Router	S0/0/0	13.13.5.2	255.255.255.252	Link to HO
SL-Router	Gi0/0	13.13.3.1	255.255.255.128	LAN Gateway
SL-Router	S0/0/1	13.13.5.6	255.255.255.252	Link to HO
SL-Router	Gi0/1	13.13.5.13	255.255.255.252	Link to Core-Switch
SD-Router	Gi0/0	13.13.3.129	255.255.255.128	LAN Gateway
SD-Router	Gi0/1	13.13.5.14	255.255.255.252	Link to Core-Switch
SD-Router	S0/0/0	13.13.5.25	255.255.255.252	Link to AU
RB-Router	Gi0/0	13.13.4.1	255.255.255.128	LAN Gateway
RB-Router	S0/1/0	13.13.5.10	255.255.255.252	Link to HO
RB-Router	Gi0/1	13.13.5.18	255.255.255.252	Link to Core-Switch
RB-Router	S0/0/1	13.13.5.29	255.255.255.252	Link to AU
AU-Router	Gi0/0	13.13.4.129	255.255.255.192	LAN Gateway
AU-Router	S0/0/0	13.13.5.26	255.255.255.252	Link to SD
AU-Router	S0/0/1	13.13.5.30	255.255.255.252	Link to RB

End Devices:

Device	IP Address	Mask	Gateway	Assignment
HO-DNSWebServer	13.13.0.2	255.255.254.0	13.13.0.1	Static
HO-EmailServer	13.13.0.3	255.255.254.0	13.13.0.1	Static
HO-PC1	13.13.0.4	255.255.254.0	13.13.0.1	Static
HO-PC2	13.13.0.5	255.255.254.0	13.13.0.1	Static
HO-Printer	13.13.0.6	255.255.254.0	13.13.0.1	Static
TC-EmailServer	13.13.2.3	255.255.255.0	13.13.2.1	Static
TC-PC1	13.13.2.11	255.255.255.0	13.13.2.1	DHCP
TC-PC2	13.13.2.12	255.255.255.0	13.13.2.1	DHCP
SL-DHCPServer	13.13.3.2	255.255.255.128	13.13.3.1	Static
SL-EmailServer	13.13.3.5	255.255.255.128	13.13.3.1	Static
SL-PC1	13.13.3.10	255.255.255.128	13.13.3.1	DHCP
SL-PC2	13.13.3.11	255.255.255.128	13.13.3.1	DHCP
SD-DHCPServer	13.13.3.130	255.255.255.128	13.13.3.129	Static
SD-EmailServer	13.13.3.133	255.255.255.128	13.13.3.129	Static
SD-PC1	13.13.3.140	255.255.255.128	13.13.3.129	DHCP
SD-PC2	13.13.3.141	255.255.255.128	13.13.3.129	DHCP
RB-EmailServer	13.13.4.3	255.255.255.128	13.13.4.1	Static
RB-PC1	13.13.4.11	255.255.255.128	13.13.4.1	DHCP
RB-PC2	13.13.4.12	255.255.255.128	13.13.4.1	DHCP
AU-EmailServer	13.13.4.130	255.255.255.192	13.13.4.129	Static
AU-PC1	13.13.4.131	255.255.255.192	13.13.4.129	Static

AU-PC2	13.13.4.132	255.255.255.192	13.13.4.129	Static
AU-Printer	13.13.4.133	255.255.255.192	13.13.4.129	Static

Router Configuration Commands

HO-Router:

```
enable
configure terminal
hostname HO-Router
interface GigabitEthernet0/0
ip address 13.13.0.1 255.255.254.0
no shutdown
exit
interface Serial0/0/0
ip address 13.13.5.1 255.255.255.252
clock rate 64000
no shutdown
exit
interface Serial0/0/1
ip address 13.13.5.5 255.255.255.252
clock rate 64000
no shutdown
exit
interface Serial0/1/0
ip address 13.13.5.9 255.255.255.252
clock rate 64000
no shutdown
exit
ip route 13.13.2.0 255.255.255.0 13.13.5.2
router rip
version 2
no auto-summary
network 13.13.0.0
network 13.13.5.0
redistribute static
exit
end
```

write memory

TC-Router:

enable
configure terminal
hostname TC-Router
interface GigabitEthernet0/0
ip address 13.13.2.1 255.255.255.0
no shutdown
exit
interface Serial0/0/0
ip address 13.13.5.2 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 13.13.5.1
ip dhcp excluded-address 13.13.2.1 13.13.2.10
ip dhcp pool TC-POOL
network 13.13.2.0 255.255.255.0
default-router 13.13.2.1
dns-server 13.13.0.2
exit
end
write memory

SL-Router:

enable
configure terminal
hostname SL-Router
interface GigabitEthernet0/0
ip address 13.13.3.1 255.255.255.128
no shutdown
exit
interface Serial0/0/1
ip address 13.13.5.6 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/1
ip address 13.13.5.13 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/0

```
ip helper-address 13.13.3.2
exit
router rip
version 2
no auto-summary
network 13.13.3.0
network 13.13.5.0
exit
end
write memory
```

SD-Router:

```
enable
configure terminal
hostname SD-Router
interface GigabitEthernet0/0
ip address 13.13.3.129 255.255.255.128
no shutdown
exit
interface GigabitEthernet0/1
ip address 13.13.5.14 255.255.255.252
no shutdown
exit
interface Serial0/0/0
ip address 13.13.5.25 255.255.255.252
clock rate 64000
no shutdown
exit
interface GigabitEthernet0/0
ip helper-address 13.13.3.130
exit
ip route 13.13.2.0 255.255.255.0 13.13.5.13
router rip
version 2
no auto-summary
network 13.13.3.0
network 13.13.5.0
exit
end
write memory
```

RB-Router:

```
enable
configure terminal
hostname RB-Router
interface GigabitEthernet0/0
ip address 13.13.4.1 255.255.255.128
no shutdown
exit
interface Serial0/1/0
ip address 13.13.5.10 255.255.255.252
no shutdown
exit
interface GigabitEthernet0/1
ip address 13.13.5.18 255.255.255.252
no shutdown
exit
interface Serial0/0/1
ip address 13.13.5.29 255.255.255.252
clock rate 64000
no shutdown
exit
ip route 13.13.0.0 255.255.254.0 Serial0/1/0
ip dhcp excluded-address 13.13.4.1 13.13.4.10
ip dhcp pool RB-POOL
network 13.13.4.0 255.255.255.128
default-router 13.13.4.1
dns-server 13.13.0.2
exit
router rip
version 2
no auto-summary
network 13.13.4.0
network 13.13.5.0
exit
end
write memory
```

AU-Router:

```
enable
configure terminal
hostname AU-Router
interface GigabitEthernet0/0
ip address 13.13.4.129 255.255.255.192
```

```
no shutdown
exit
interface Serial0/0/0
ip address 13.13.5.26 255.255.255.252
no shutdown
exit
interface Serial0/0/1
ip address 13.13.5.30 255.255.255.252
no shutdown
exit
ip route 13.13.0.0 255.255.254.0 13.13.5.29 200
router rip
version 2
no auto-summary
network 13.13.4.0
network 13.13.5.0
exit
end
write memory
```

Assumptions

1. A Core Switch is used to interconnect SL, SD, and RB routers using GigabitEthernet links since they are described as interconnected through a core switch in the topology requirements.
2. All inter-router serial links use /30 subnets allocated from the 13.13.5.0/24 address space which is separate from the unit subnets.
3. Email domains were changed from .sisn.local to .sisn.com because Cisco Packet Tracer does not support .local as a valid email domain format.
4. DHCP excluded addresses reserve the first 10 IPs in each pool to avoid conflicts with statically assigned servers and routers.
5. HO and AU devices are all configured with static IP addresses as required by the project instructions.
6. TC and RB routers act as DHCP servers directly since no dedicated DHCP server device is required for those units.
7. SL and SD use dedicated server devices for DHCP with ip helper-address configured on their respective routers to forward DHCP broadcasts.
8. The floating static route on AU-Router uses Administrative Distance of 200 which is higher than RIPv2's default AD of 120 ensuring the static route only activates when the RIP path via SD fails.
9. HO redistributes static routes into RIPv2 so all dynamically routing units can learn the route to TC.